



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Beauty

Lesson 10-2 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can use bilateral symmetry to describe structure in nature and in human creations.

Language Objective: I can describe a symmetric design using the words mirror image, line of symmetry, and balance.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

symmetric

mirror image

bilateral symmetry

line of symmetry

balance



Tap any word to see what it means and a picture.

1

Having two halves that match as mirror images is being

2

A copy of a figure flipped across a line is its

3

Reflective symmetry in plants and animals is called

4

The line you draw so each side reflects the other is the

5

An even arrangement where neither side feels heavier is

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How do you respond to your friend?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **symmetric** — Having two halves that match — one side is a mirror image of the other.

simétrico — Que tiene dos mitades que coinciden: un lado es la imagen reflejada del otro.

- ☐ **mirror image** — A copy of a figure flipped across a line, so the two halves face each other.

imagen reflejada — Una copia de una figura volteada al otro lado de una línea, de modo que las dos mitades se miran entre sí.

- ☐ **bilateral symmetry** — Reflective symmetry in plants and animals — the two sides of the body mirror each other.

simetría bilateral — La simetría reflectiva en plantas y animales: los dos lados del cuerpo se reflejan entre sí.

- ☐ **line of symmetry** — The line you can draw through a figure so each side is a reflection of the other.

eje de simetría — La línea que puedes trazar por una figura de modo que cada lado sea el reflejo del otro.

- ☐ **balance** — An even, steady arrangement — neither side feels heavier than the other.

equilibrio — Una disposición pareja y estable: ningún lado se siente más pesado que el otro.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

To settle the disagreement, draw a ____ through the center of each flower and check whether one side is a ____ of the other. If the test works for both, then ____ flowers have bilateral symmetry.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A figure is symmetric only if there is a specific line where one side is a mirror image of the other.

Evidence: A strong answer explains the line gives you something to compare the two sides across — without it, you can't test whether the sides actually match. A weak answer just says 'it looks the same.'

Reasoning: This evidence connects the definition of symmetric to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- To settle the disagreement, draw a ____ through the center of each flower and check whether one side is a ____ of the other. If the test works for both, then ____ flowers have bilateral symmetry.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- I know because ____.
Lo sé porque ____.
- So ____.
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** symmetric
2. **Notes 2:** mirror image
3. **Notes 3:** bilateral symmetry
4. **Notes 4:** line of symmetry
5. **Notes 5:** balance
6. **Watch for this mistake:** *Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical — the mirror line can run in any direction, and the test is whether the two sides match across it.*
7. **Try It 1:** A. 46 flowers — *One side uses $14 + 9 = 23$ flowers, and the mirror side uses the same 23, so the whole design uses $23 \times 2 = 46$ flowers.*
8. **Try It 2:**